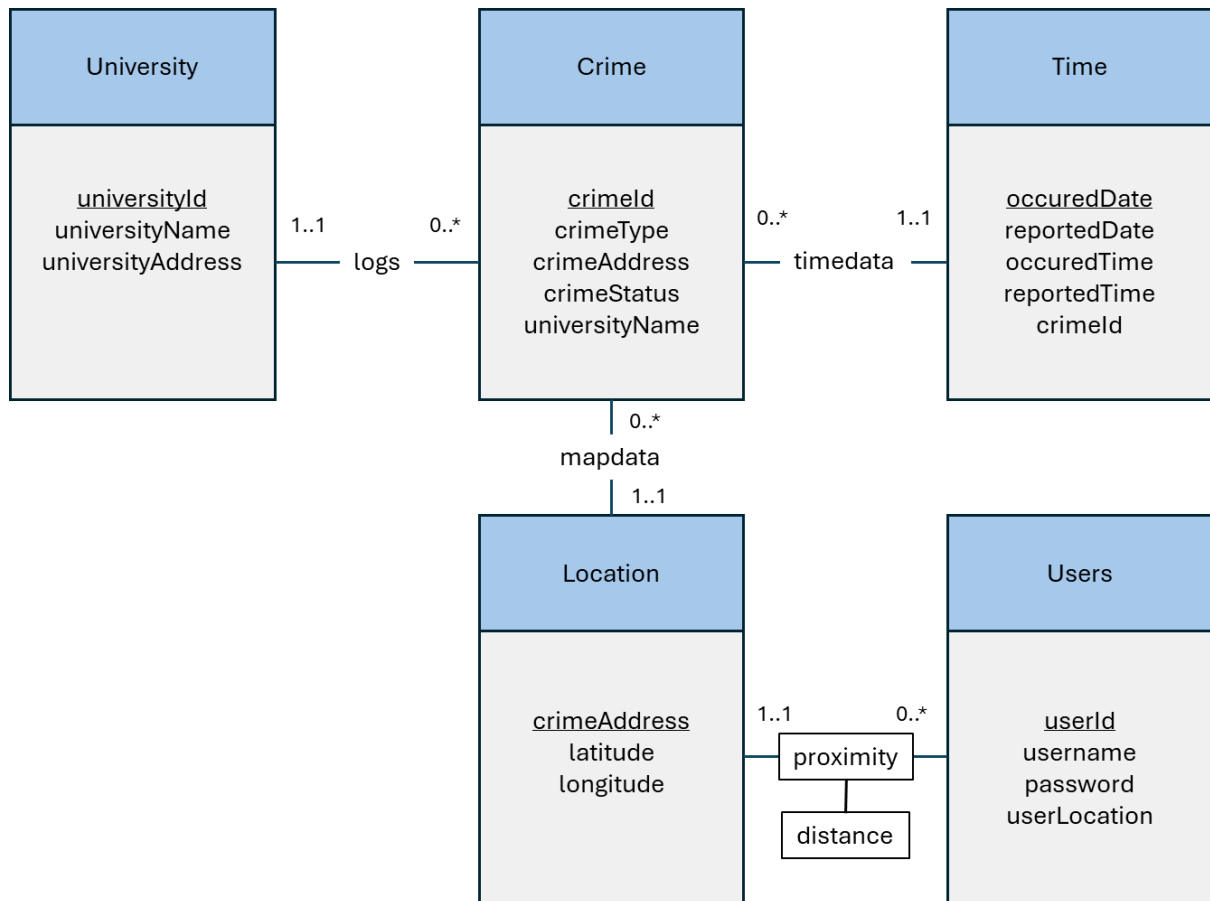




Assumptions

Entities and Attributes

1. **Crime:**
 - Attributes: crime_id (PK), status, crime_type, address, university
 - Each crime is identified by a unique ID and contains a status, location, and type.
2. **Location:**
 - Attributes: address (PK), latitude, longitude
 - Each location is identified by a unique ID and contains an address where the crime occurred.
3. **Time:**
 - Attributes: occurredDate (PK), reportedDate, occurredTime, reportedTime
 - Each crime is identified by a unique ID and contains the specific date of the crime.
4. **University:**
 - Attributes: university_id (PK), name, address
 - Each university is identified by a unique ID and contains the name and address of the university.
5. **Users:**

- Attributes: `user_id` (PK), `location_id`(FK), `username`, `password`
- Each user is identified by a unique ID and contains the username, location, and the password..

Relationships and Cardinality

1. **Crime - Location (1,∞):**
 - Each crime occurs at one specific location.
 - One location can have multiple crimes.
2. **Crime - Date (1,∞):**
 - Each crime occurs on a specific date.
 - One date can have multiple crimes.
3. **Crime - University (1,∞):**
 - Each crime occurs at one university
 - One university can have multiple crimes
4. **User - Location (1, ∞):**
 - Each user should only have 1 location
 - Each location can have multiple users

Normalization:

FDs:

`crimeId`-> `crimeType`,`crimeAddress`,`crimeStatus`,`universityName`
`crimeId`->`occurredDate`,`reportedDate`,`occurredTime`,`reportedTime`
`universityId`->`universityName`,`universityAddress`
`crimeAddress`->`latitude`,`longitude`
`userId`->`username`,`password`,`userLocation`

`Crime`(`crimeId`,`crimeType`,`crimeAddress`,`crimeStatus`,`universityName`)
`Time`(`crimeId`,`occurredDate`,`reportedDate`,`occurredTime`,`reportedTime`)
`University`(`universityId`,`universityName`,`universityAddress`)
`Location`(`crimeAddress`,`latitude`,`longitude`)
`Users`(`userId`,`username`,`password`,`userLocation`)

Relational Schema

1. University(universityId:INT [PK], universityName:VARCHAR(X), universityAddress:VARCHAR(X))
2. Crime(crimeId:INT [PK], crimeType:VARCHAR(X), crimeAddress:VARCHAR(X), crimeStatus:VARCHAR(X))
3. Location(locationId:INT [PK], crimeAddress:VARCHAR(X), latitude:REAL, longitude:REAL)
4. Time(occurredDate:DATE [PK], reportedDate:DATE, occurredTime:VARCHAR(X), reportedTime:VARCHAR(X),), crimeId:INT)
5. Users(userId:INT [PK], username:VARCHAR(X), password:VARCHAR(X), userLocation:VARCHAR(X))
6. MapData(crimeAddress:VARCHAR(X), crimeId:INT)
7. TimeData(crimeId:INT)
8. Proximity(userId:INT, crimeAddress:VARCHAR(X), distance:REAL)